

BLOCK 4

Production & Homework Launch

Part 1: Production readiness · Part 2: Brief + Design session

Block 4 — 3 hours

The demo-to-production gap

What works in the lab will break in production. Why. The checklist.

Monitoring

5 metrics that detect 90% of production failures.

Versioning

Hash the system prompt. Trace every behaviour change.

EU AI Act

4 risk tiers. Where your agent sits. What obligation follows.

Production checklist

Security · Reliability · Retrieval · Reasoning · Compliance.

Homework brief

Task, topic list, deliverable format, submission rules.

Design session

Groups · topic · problem statement · architecture · PEAS · GitHub repo.

01

Production Readiness

From the lab to the first real user

The demo-to-production gap

Works in the lab	Breaks in production	Fix required
3 test documents	300 documents, edge cases, ambiguous queries	RAGAS at scale + chunking validation
1 user, 1 query	100 concurrent users, 100 different query types	Rate limiting + queue management
Happy path only	Tool timeouts, API errors, budget overruns	Retry + circuit breaker on all tools
Known input format	Malformed inputs, injection attempts	L1 filter (you built this)
Answer always available	New questions, insufficient data	'Insufficient data' path + CONFIDENCE flag
No monitoring	Silent failures, cost drift, quality degradation	Langfuse alerts + weekly RAGAS evaluation
Works on your machine	Fails from a clean clone	README + requirements.txt + .env.example — tested

EU AI Act — Where your project sits

PROHIBITED	Social scoring · unconscious manipulation · real-time biometric surveillance · predictive policing by profiling. Do not build.	<i>In force Aug 2024</i>
HIGH RISK	HR decisions · credit scoring · educational assessment · law enforcement · critical infrastructure · migration. HITL + audit trail + conformity assessment mandatory.	<i>Required Aug 2026</i>
LIMITED RISK	Chatbots · AI-generated content · research tools · recommendation systems (public-facing). Transparency obligation: inform users they are interacting with AI.	<i>Likely your project</i>
MINIMAL RISK	Spam filters · internal research tools · AI in games. No mandatory regulatory requirements.	<i>No obligations</i>

For your report: identify your tier with a one-paragraph justification. If your output influences decisions about real individuals → High Risk.

Production checklist — before the first real user

Security	Reliability	Retrieval	Reasoning	Compliance
☐ L1 filter: 5/5 injection tests passed	☐ All tools return strings (no exceptions)	☐ Hybrid search (BM25 + dense + RRF)	☐ Few-shot CoT (EVIDENCE/ANALYSIS/CONCLUSION/CONFIDENCE)	☐ EU AI Act tier assessed and documented
☐ L4 gate: SAFE/MONITOR/CONFIRM/BLOCK for all tools	☐ max_steps limit in agent loop	☐ Cross-encoder reranking applied	☐ Self-Consistency $k \geq 3$ on final synthesis	☐ User disclosure implemented
☐ Token budget: max_usd set + recorded after each call	☐ finish_reason='length' handled	☐ RAGAS baseline + improved documented	☐ Failure mode constraints in system prompt	☐ Data retention policy defined

02

Homework Brief

Topic · deliverable · design session · deadline

The task — build a complete agent

①	Production RAG pipeline	Hybrid search + cross-encoder reranking + RAGAS documented. Baseline vs improved.
②	Custom MCP server	3+ tools with complete docstrings. Error handling on every tool. 3/3 inspector tests passed.
③	Security stack	L1 filter + L4 gate + token budget. 5/5 injection tests passed. Tests in tests/test_security.py.
④	Reasoning strategy	Few-shot CoT with EVIDENCE/ANALYSIS/CONCLUSION/CONFIDENCE. Self-Consistency $k \geq 3$.
⑤	Critic agent	At minimum: Generator-Critic pattern. The critic produces a visible verdict before returning the answer.
⑥	REPORT.md	Problem statement · architecture · RAGAS table · security before/after · EU AI Act · limitations · AI use declaration.

Design session — before you leave

①	Form your group	2–3 students. Register group number and topic with the instructor.
②	Choose your topic	Topic list in the homework brief. First registered, first served.
③	Problem statement	One paragraph: who is the user, what the agent does that a chatbot cannot, one concrete scenario.
④	Architecture + PEAS	Agent loop, tools (name + what they return), vector store content, MCP server. PEAS: Performance (specific!), Environment, Actuators, Sensors.
⑤	Create the GitHub repo	Required structure: src/ tests/ docs/ data/ README.md REPORT.md .env.example requirements.txt